IMAD ABOO

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EDUCATION

General Engineering (Intended B.S. in **Mechanical Engineering**) | **Virginia Tech** Aug 2026 – May 2030
High School Diploma '26 | **Acton-Boxborough Regional High School** Sept 2022 - May 2026

- **3.739** GPA with **All Honors** + **AP** Coursework (Including but not limited to AB Calculus, Statistics, Physics, etc)

EXPERIENCE

Airframe-AE-Array July 2026

- Co-built **software** simulating structural cracks in aircraft via real-time **acoustic-emission** sensing
- Co-wired piezoelectric microphones to an **Arduino** Uno Q to **physically triangulate impact**
- Built the **hardware-to-software** integration using **Python** for the acoustic-emission sensor array, **connecting** piezoelectric **sensor** input to the **simulation** visualization

Windmesh May 2026

- Led a 4-person team, during which we utilized **vector-calculus**, **OpenCV**, and the **Rothermel fire spread model** to design and create software used to predict fire spread
- **Integrated hardware-to-software** using **Python**, feeding live anemometer windspeed data into a simulation node to model real-time fire spread response to wind conditions
- Spearheaded summary **video & presentation** for local hackathon "LexHacks", earning **2nd place**

Inspirit AI July 2025

- Built and trained a **convolutional neural network** (CNN) using TensorFlow, Keras, and Pandas to classify pneumonia from chest X-ray images, achieving **92%** accuracy as part of a small cohort

Student Participant: SummerLab | CityLab at Boston University June 2025

- Performed **CRISPR gene-editing** experiments and was among the first to use a novel **dCas-9** strand; fluorescent results were adopted by the **Stanley Lab** in **UC Berkeley** to advance development of a new dCas-9 variant.
- Engaged with lab equipment and became proficient in various lab techniques (micro-pipetting, gel electrophoresis, cell culturing, PCR, etc)

LEADERSHIP

President, Muslim Student Association Sept 2022 - May 2026

- Progressed from social media manager to **President** of **16-member** club
- Led initiatives amid adversity, using curated media & **8+** annual events (henna sessions, bake sales, student panels, etc) to build community & normalize Muslim belonging

Historian, Speech & Debate Sept 2022 - May 2026

- Ranked **27th** statewide out of **100+** competitors in Varsity Extemporaneous Speaking
- Recruited & mentored **4** novice members and as Historian grew the team's social media following by **~25%** through tournament recap content, supporting a **31**-member club

ADDITIONAL INFORMATION

- **Technical Skills**: Python, TensorFlow, Keras, Pandas, OpenCV, Roboflow, Arduino, Github, General lab techniques
- **Soft Skills**: Leadership, Team collaboration, Public speaking, Mentorship, Adaptability
- **140+** Volunteer Hours: Emerson Hospital, School volunteering, & Volleyball
- **Languages**: English, Spanish, Tamil, Arabic
- **Certifications**: The Massachusetts State Seal of Biliteracy (Spanish)